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9.3.2 The Power Method: Convergence

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particular formulation tells us is

that eventually all of these terms become arbitrarily small, because λ_0 is the eigenvalue largest in magnitude, which means that we expect to eventually point in the direction of ψ_0 times x_0 . In other words in the direction of x_0 except the vector that will compute will be scaled by the coefficient the first coefficient of our vector y . So this now allows us to see how fast we actually

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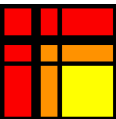
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Reading assignment

1/1 point (graded)



Read Unit 9.3.2 of the notes. [\[LINK\]](#)

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Homework 9.3.2.1

1/1 point (graded)

Let $\mathbf{X} \in \mathbb{C}^{m \times m}$ be nonsingular. Define $\|\cdot\|_{\mathbf{X}^{-1}} : \mathbb{C}^m \rightarrow \mathbb{R}$ by $\|\mathbf{y}\|_{\mathbf{X}^{-1}} = \|\mathbf{X}^{-1}\mathbf{y}\|$ for some given norm $\|\cdot\| : \mathbb{C}^m \rightarrow \mathbb{R}$.

ALWAYS/SOMETIMES/NEVER: $\|\cdot\|_{\mathbf{X}^{-1}}$ is a norm.

ALWAYS

✓ Answer: ALWAYS

Solution:

For a complete solution see the [notes](#).

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